YojanaBench — Setup Guide (Step by Step, Very Simple)

Follow these steps in order. If you get stuck, jump to the **Troubleshooting** section at the end. Total time: about 10–15 minutes.

> *This project runs **100% on your own laptop**. No database, no sign-up, no API keys, no paid services. You only need **Python** and a few **libraries**.*

Part 0 — What you need before starting (Requirements)

Please make sure your laptop has:

- **A computer** running Windows, macOS, or Linux.
- **Python version 3.10 or newer** (3.10, 3.11, 3.12 all work). We will check this in Step 1.
- **pip** — the tool that installs Python libraries. It comes automatically with Python, so you usually get it for free.
- **A text editor** — we recommend **VS Code** (free): <https://code.visualstudio.com>
- **About 500 MB** of free disk space.
- **Internet access for the first setup only** (to download the libraries). After that, the project runs offline.
- Optional: **Git** (only if you want to download the project with a `git clone` command instead of copying the folder).

You do **not** need:

- A database (data is stored in simple text/JSON files).
- Any AI account or API key.
- Docker or any cloud service.

Part 1 — The libraries we need (what `requirements.txt` means)

The file `requirements.txt` in the project folder lists the libraries. Here is what each one does, in simple words:

- **fastapi** — the web framework that runs the backend (the "engine room").

- **uvicorn** — the small server that actually starts FastAPI and listens for requests.
- **requests** — a helper for making web requests (used by a few background tasks).
- **pydantic** — checks and cleans the data that comes in and goes out.
- **deep_translator** — translates text for the multilingual voice assistant. *Optional* — the app still works in English without it.
- **pytest** — runs the automatic tests. *Optional* (only for testing).
- **httpx** — a helper that the tests use to call the API. *Optional* (only for testing).

So there are **4 must-have** libraries and **3 optional** ones. Installing all of them is easiest, and that is exactly what one command does (Step 5).

Part 2 — Step-by-Step Setup

Step 1 — Install Python (if you don't have it)

Check first. Open a terminal and type:

```
python --version
```

- If you see something like `Python 3.10.9`, you are ready — go to Step 2.
- If you see an error like *"python is not recognized"*, install Python:
- Go to <https://www.python.org/downloads/> and download Python 3.10 or newer.
- **Windows important:** on the installer's first screen, tick the checkbox **"Add Python to PATH"** before clicking Install. This is the most common mistake.
- **macOS:** download the installer and run it. Or, if you use Homebrew: `brew install python`.
- Close and reopen the terminal, then run `python --version` again.

Step 2 — Get the project files

Option A (simplest): copy or unzip.

Copy the whole `YojanaBench` folder to your laptop, for example to `C:\Users\YourName\YojanaBench` (Windows) or `~/YojanaBench` (Mac/Linux).

Option B (if you have Git):

```
git clone <repository-url>
```

Either way, you should now have a folder named `YojanaBench`.

Step 3 — Open a terminal inside the project folder

You must run the next commands **from inside the YojanaBench folder**.

- **Windows:** open the `YojanaBench` folder in File Explorer, click the address bar, type `powershell`, and press Enter. A terminal opens already inside the folder.
- **VS Code:** open the folder in VS Code, then use the menu **Terminal** → **New Terminal**. It starts inside the folder.
- **macOS/Linux:** open Terminal and type `cd` followed by the folder path, for example `cd ~/YojanaBench`.

Check you are in the right place by listing the files:

```
dir        # Windows (PowerShell)
ls         # macOS / Linux
```

You should see `backend`, `frontend`, `data`, `run.py`, `requirements.txt`, etc.

Step 4 — (Recommended) Create a private workspace ("virtual environment")

A **virtual environment** keeps this project's libraries separate from the rest of your computer. It is optional but strongly recommended.

Create it (once):

```
python -m venv .venv
```

Activate it:

```
.\.venv\Scripts\Activate.ps1     # Windows (PowerShell)
source .venv/bin/activate        # macOS / Linux
```

When it is active you will see `(.venv)` at the start of every terminal line.

> *If Windows blocks activation with a message about *"running scripts is disabled"*, run this once, then try again:*
>
> `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`

Step 5 — Install the libraries (the main "import" step)

This single command reads `requirements.txt` and installs everything:

```
pip install -r requirements.txt
```

What this means: **pip** = Python's installer; **install** = download and set up;
-r requirements.txt = "read the list of libraries from this file".

> *If* `pip` *is not recognized, try* `python -m pip install -r requirements.txt`*.*
>
> *If it fails on Windows, close and reopen the terminal first.*

To install **only the must-have libraries** (a lighter option), you can instead run:

```
pip install fastapi uvicorn requests pydantic
```

Step 6 — Verify that everything installed correctly

List the installed libraries:

```
pip list
```

You should see `fastapi`, `uvicorn`, `pydantic`, and `requests` in the list.

You can also do a quick import test:

```
python -c "import fastapi, uvicorn, pydantic, requests; print('All good!')"
```

If you see `All good!`, the setup is complete.

Step 7 — Start the project

Run the one launcher command:

```
python run.py
```

You should see a banner, then a line like
`Uvicorn running on http://127.0.0.1:8000`. The launcher also tries to open your browser automatically. **Leave this terminal open** — it is the running server.
Press **Ctrl + C** in the terminal to stop it later.

Step 8 — Open it in the browser

If the browser did not open by itself, go to:

- **Dashboard (login first):** <http://127.0.0.1:8000>

- **Grievance & Money-Trace portal:** <http://127.0.0.1:8000/grievance>
- **API documentation:** <http://127.0.0.1:8000/docs>

On the dashboard you will see a **login screen first**. Use these demo accounts:

- Officer: `officer / officer123` (or `admin / admin123`)
- Citizen whose money was stolen: `geeta / geeta123`
- Citizen whose money arrived correctly: `ravi / ravi123`

Step 9 — (Optional) Run the automatic tests

This proves the code is working on your machine:

```
python -m pytest -q
```

You should see a line like `103 passed`. This is optional but a good confidence check.

Part 3 — Daily use (after the first setup)

Once installed, starting the project each day is just three lines:

```
.\.venv\Scripts\Activate.ps1     # activate the workspace (Windows)
python run.py                    # start the server
# then open http://127.0.0.1:8000 in your browser
```

(On macOS/Linux use `source .venv/bin/activate`.)

Part 4 — Troubleshooting (common problems)

1. `python is not recognized` **/** `pip is not recognized`

Python is not installed, or was installed without "Add to PATH". Reinstall Python and tick **"Add Python to PATH"**. Then reopen the terminal. Try `python -m pip` instead of plain `pip`.

2. `ModuleNotFoundError: No module named 'fastapi'` **(or similar)**

The libraries are not installed, or your virtual environment is not active. Activate it (Step 4) and run `pip install -r requirements.txt` again.

3. Running scripts is disabled on Windows (cannot activate `.venv`**)**

Run this once, then retry activation:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

4. `[Errno 98] Address already in use` **or the page shows old content**

Another server is already using port 8000 (maybe an old run). Close the old terminal, or find and stop the process, then run `python run.py` again. On Windows:

```
Get-NetTCPConnection -LocalPort 8000 | ForEach-Object { Stop-Process -Id $_.OwningProcess -Force }
```

5. The page looks outdated / a change is not showing

The browser cached the old page. Do a **hard refresh** with **Ctrl + F5** (Windows) or **Cmd + Shift + R** (Mac).

6. The dashboard loads directly without asking to log in

You were already signed in (the session is remembered). Click **Sign out** in the top-right, or clear the site data, and the login screen will appear.

7. The voice assistant does not speak

It is optional. The app works fine without it. Make sure you are using a modern browser (Chrome/Edge) and allow sound.

8. Something else is broken

Re-run the tests (`python -m pytest -q`) and read the first error message — it usually names the exact missing piece.

Part 5 — Uninstall / clean up

To remove the project's libraries, just delete the `.venv` folder inside the project (if you created one). To remove the project, delete the whole `YojanaBench` folder. Nothing else on your computer is changed.

Part 6 — Quick reference card (one screen)

```
# 1. go into the project folder
cd C:\Users\YourName\YojanaBench

# 2. create + activate a workspace (first time only)
python -m venv .venv
.\.venv\Scripts\Activate.ps1
# 3. install the libraries (first time only)
```

```
pip install -r requirements.txt

# 4. run the project
python run.py

# 5. open in the browser
#    http://127.0.0.1:8000          (dashboard - login first)
#    http://127.0.0.1:8000/grievance (money-trace portal)
```

Demo logins: `officer / officer123`, `geeta / geeta123`, `ravi / ravi123`.